

The Case for Ito

While the Stratonovich integral has a certain geometric elegance (and obeys the standard chain rule we have all learned), the Ito integral is the backbone of modern finance and filtering for three very good reasons:

- **Non-Anticipatory Nature:** In Ito calculus, the integrand is evaluated at the beginning of the time interval. This aligns well with reality: you cannot make an investment decision based on where the price is going to be at the end of the interval. It respects the arrow of time.
- **The Martingale Property:** This is the main feature. The Ito integral of a process with respect to Brownian motion is a martingale. This makes it very easy to calculate expectations ($E[X_t]$), which is why it's the gold standard for option pricing (Black-Scholes).
- **Computational Simplicity:** Because the noise is independent of the current state, numerical schemes are straightforward to implement.
- **Visualizing the Difference:** To see the difference, we can look at a Stochastic Differential Equation (SDE) where the noise depends on the state (multiplicative noise).

A typical example is:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

The key difference is that the Stratonovich interpretation effectively adds a drift term that isn't present in the Ito version. Specifically, the Stratonovich SDE $\circ dX_t = \sigma X_t \circ dW_t$ is equivalent to the Ito SDE $dX_t = \frac{1}{2}\sigma^2 X_t dt + \sigma X_t dW_t$

Sample Paths

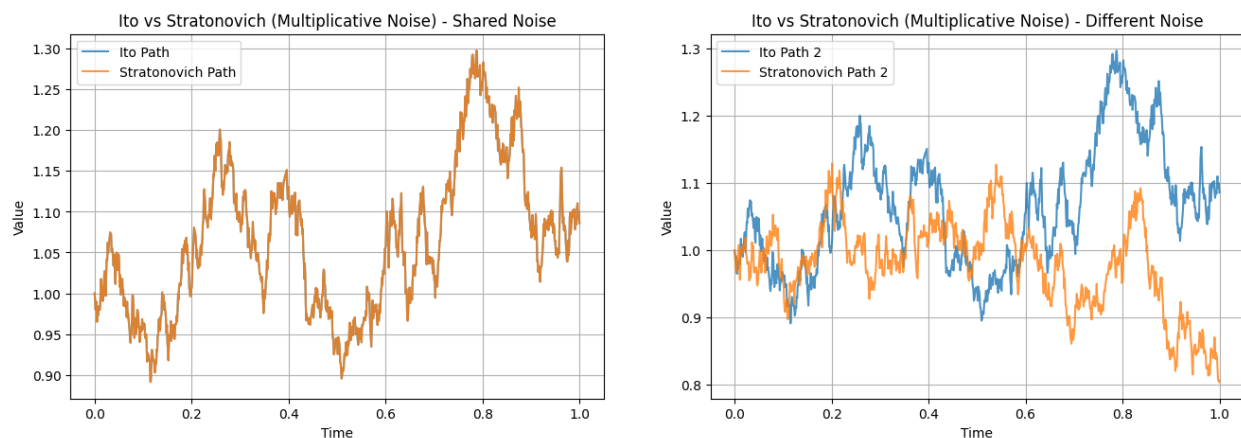


Figure 1: Samples Paths: Ito vs Stratonovich (Multiplicative Noise)